Homework 03: Probability and Random Variables

Problem 1: A Gamble

We are gambling. You roll one six-sided die.

- if 1 or 2 or 3 is rolled, you win $\$0$,
- if 4 or 5 is rolled, you win $\$3$,
- if a 6 is rolled, you lose $\$-7$.

Define the random variable

$$X = \text{the amount you win in dollars}$$

- a. Write out the simplest (with respect to events) probablility space that models this gamble. If you had to write all the possible events, what would the answer look like?
- b. Write the PMF of X and compute $\mathbb{E}[X]$.
- c. Would a player who only cares about expected value want to play this game?

Problem 2: Weather Forecasting

A weather forecaster has the following track record:

- on 30% of days it rains,
- when it rains, they predicts rain 80% of the time,
- when it does not rain, they predicts rain 10% of the time.

If they predict rain today, what is the probability that it actually rains?

Problem 3:

You roll two fair, six-sided dice. For the following random variables, compute the probability mass function and expectation:

- a. The average of the two die.
- b. The minimum of the two die.
- c. The maximum of the two die.
- d. Clearly explain the difference between the probability space, the random variable, and the probability mass function.

Problem 4: Defective Parts

Parts are produced at two factories, A and B. A proportion r of parts come from factory A, and $1-r$ come from factory B.

Factory A produces defective parts with probability a, and factory B produces defective parts with probability b.

If a defective part is discovered, it is not obvious whether it came from A or B, but the repair process depends on the source.

1. Make a probability tree that starts with the factory source and then resolves whether the part is defective.
2. What is the probability that a randomly selected part is defective?
3. If a part is defective, what is the probability that it came from factory A?